

FresnelDeformable: A Softness Presentation Technique Combining Fingertip Plane Tilt Manipulation and Pseudo-Stiffness

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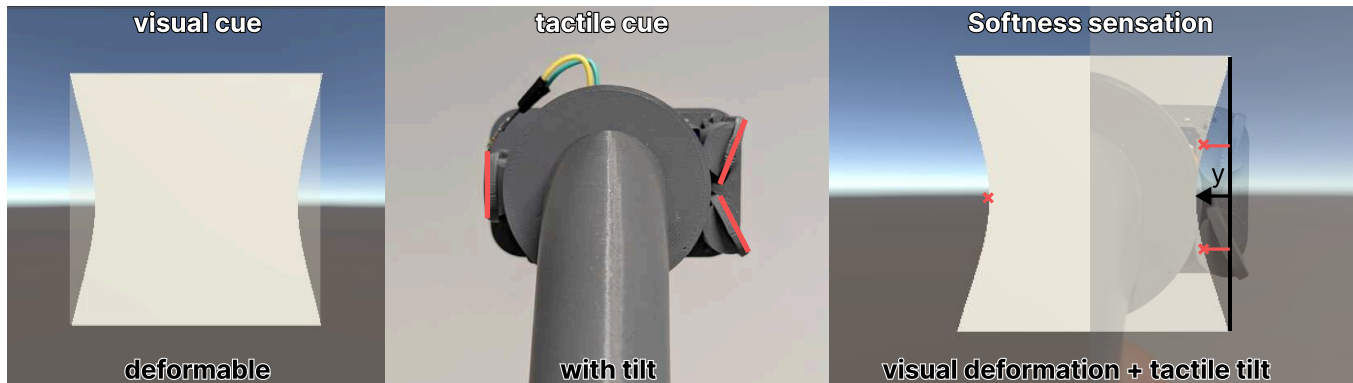


Figure 1: FresnelDeformable is a system that conveys softness perception by leveraging visual cues (the deformation of a virtual object) and tactile cues (the orientation of each fingertip contact plane of our custom haptic device).

Abstract

We present *FresnelDeformable*, a novel approach for conveying the softness of virtual objects by combining pseudo-stiffness with dynamic fingertip plane tilt manipulation. Our method leverages a custom device equipped with servo motors that tilt each fingertip contact surface according to the user's applied force, aligning the tactile feedback with the visually simulated deformation of a virtual object. In a pilot user study, we compared three types of visual stimuli (compressible, deformable, and no-deformation) and two

tactile conditions (with and without tilt) to evaluate their effects on perceived softness, realism, and visuo-tactile coherence. The results showed that tilt manipulation significantly increased the coherence between visual and tactile cues, and also suggested a moderate-to-large effect on softness perception. These findings imply that orienting the fingertip contact plane can complement existing pseudo-stiffness techniques, expanding the range of perceived softness in virtual environments.

CCS Concepts

• **Hardware** → *Emerging interfaces*; • **Human-centered computing** → *Haptic devices*.

Keywords

Pseudo-haptics, Softness, Handheld device

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1 Introduction

In recent years, how best to convey the sense of softness of objects in interactive environments, such as virtual reality (VR) and augmented reality (AR), has become an important research topic [13]. Many everyday objects undergo actual deformation when pressed or gripped, and the tactile sensations arising during these interactions serve to communicate the objects' physical characteristics. At the same time, people also rely on visual cues—such as seeing an object's surface compress or bend—to infer how soft it is. Likewise, in virtual objects, the perception of softness is a composite sensation produced by integrating multiple haptic cues, such as “compressibility” and “elasticity,” together with corresponding visual feedback [5]. However, realizing softness perception purely through mechanical means often requires large and complex devices, resulting in bulky systems. Against this backdrop, there has been active research on simpler and more lightweight methods for softness presentation, leveraging the interplay between visual and tactile information [5, 10, 16].

One such approach is known as “pseudo-stiffness,” in which even a rigid, non-deformable object can appear soft or stiff by visually depicting a deformation animation. Previous studies have reported that adding vibration [5] or passive haptic cues [1, 8, 17, 18] can enhance the natural acceptance of these perceived stiffness or softness sensations. However, existing pseudo-stiffness methods typically rely on a one-degree-of-freedom spring model and do not sufficiently account for dynamic changes in the orientation or deformation of the surface with which the fingertip makes contact.

Meanwhile, it is known that the orientation of the fingertip contact plane plays a crucial role in estimating an object's curvature [9, 14]. Instead of merely showing a visual animation of pressing into a surface, dynamically altering the angle of the contact surface where the fingertip touches could affect the distribution of forces on the skin, potentially inducing a more natural softness perception. In this study, we leverage changes in the orientation of the contact surface as a key factor in the integration of visual and tactile cues, and propose a novel method for softness presentation (Fig. 1).

As a pilot study, we conducted an evaluation experiment with 10 participants under a total of six conditions, combining three types of visual stimuli with either the presence or absence of orientation-based tactile cues. Using Softness, Realism, and Visuo-tactile Coherence as primary measures, we found that orientation-based cues significantly improved the coherence between visual and tactile perceptions and showed potential to influence softness perception. On the other hand, no statistical interaction was detected with the type of visual stimulus, suggesting that the effect of changing the orientation of the contact surface might be dominant on its own.

This study makes two primary contributions: (1) it proposes a new method for conveying softness by focusing on changes in the orientation of the contact surface at the fingertip, and (2) it empirically investigates how these orientation changes affect perceived softness of virtual objects.

2 Related Work: Pseud-Stiffness and Passive Haptics

Techniques that leverage visual–haptic interactions to present the tactile properties of objects have garnered significant attention [13]. One such approach is known as pseudo-haptics, and when used specifically to convey the stiffness of an object, it is referred to as pseudo-stiffness. Stiffness corresponds to the perceived extent to which an object or surface deforms under an applied force [16], thus strongly influencing the user's perception of “hardness” or “softness” in virtual objects.

Pseudo-stiffness is typically implemented by associating the user's force input or positional input with the visual deformation of a virtual object, adjusting the ratio between them (i.e., the control–display ratio). For example, TORC [11], Selfrionette [8], and [4] deform a virtual object in response to force input, thereby conveying pseudo-stiffness. Specifically, TORC [11] visually alters a small, pinchable virtual object based on the user's applied force, enabling a sense of grasping a deformable object. In contrast, [4] demonstrated that, even without a noticeable change in the fingertip's position, convincing stiffness could be presented for objects exceeding 24 N/cm in stiffness. Moreover, Selfrionette [8] formalized the force–deformation relationship via a dynamic differential equation, suggesting the possibility of presenting tactile sensations that extend beyond simple spring-like stiffness. Together, these studies establish the significance of conveying an object's stiffness without requiring large fingertip displacements. However, as [4] notes, such an approach is currently limited to objects stiffer than approximately 24 N/cm when no real deformation occurs, and often necessitates visually exaggerated deformations. Psychophysical experiments show that a visual deformation of around 19.6% (relative to a reference compliance) is required to perceive a change in compliance under pseudo-haptics [7], while other studies have reported a necessary visual deformation of about 14–25% [16].

Another strategy for stiffness presentation uses simple tangible objects to enhance pseudo-haptics, referred to here as passive haptics, particularly when a spring-like object is employed to transform position and force without actuators. By providing a baseline compliance as real-world haptic feedback, passive haptics promotes acceptance of visually induced deformations [10]. Examples include Virtual Mitten [2] and FlexiFingers [1], both of which implement pseudo-stiffness through passive haptics. Virtual Mitten [2] maps the compression of an opposed spring to the gripping force of a virtual hand, enabling interactions like squeezing fruit. FlexiFingers [1], an exoskeleton-style device fitted with springs on each finger, enhances perceived realism when grasping virtual objects ranging from soft sponges to firm apples. Furthermore, passive haptics can also be applied to real objects. For instance, SoftAR [12] projects deformations onto a plush toy in accordance with the user's hand motions, presenting a level of softness that surpasses the toy's intrinsic properties. These studies suggest that combining visual

cues with appropriate haptic feedback can significantly expand the range of perceivable stiffness using relatively simple mechanical setups.

Combining multiple cues can also be observed in work of augmented pseudo-stiffness with transient vibration [5]. While their findings indicate that visual stimuli remain the dominant factor in softness perception, adding vibration extended the range of perceivable softness.

Taken together, these research suggest that blending visually driven pseudo-stiffness with simple yet effective haptic cues can present a broader range of softness than conventional approaches relying on only visual feedback and user-applied force. Although prior studies have widely acknowledged that visually simulating surface deformation effectively supports stiffness or softness perception, they have not closely investigated how to integrate such visual cues with tactile stimuli that also mimic surface deformation. Notably, earlier research [14] indicates that the orientation of the fingertip contact plane plays a more decisive role in curvature perception than fingertip position. Building on these insights, the present work explores whether augmenting pseudo-stiffness with a dynamic presentation of the fingertip contact surface’s orientation—thereby simulating surface deformation—can further enhance the perception of softness.

3 User Study: Perception of Softness through the Interaction of Visual and Haptic Factors

3.1 Hypotheses

The aim of this experiment is to clarify how combining the visual deformation of a virtual object with tilt manipulation at each fingertip contact surface influences perceived softness. We propose the following hypotheses:

- **H1:** By manipulating the tilt of the fingertip contact surfaces, softness can be presented even without visual deformation.
- **H2:** The combination of fingertip tilt manipulation and pseudo-stiffness enhances the overall ability to convey softness.

The first hypothesis posits that changing the orientation of the fingertip contact surfaces itself, independent of visual cues, can induce a sense of softness. Because our proposed method adjusts the tilt of each fingertip contact surface in response to force input, it is possible that even if the virtual object does not visually deform, the user’s fingertip force distribution changes sufficiently to evoke a softness perception.

The second hypothesis forms the core of this study. When visual deformation is supplemented by tactile feedback that is consistent with those visual cues, the user’s perception of softness may be further enhanced.

3.2 Experimental Set Up and FresnelDeformable

Figure 2 shows the appearance of the device used in our experiment. This device is equipped with a force sensor, three servo motors, and fingertip contact surfaces whose orientation can be controlled by the servo motors. The force sensor (FSR402) measures the user’s grip force when the device is held. Furthermore, the three servo motors (Feetech FS90), driven by an M5Stack ATOM MOTION module (which incorporates an M5 ATOM S3), dynamically manipulate the

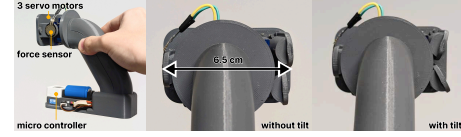


Figure 2: The fingertip contact surface orientation provided by the experimental device called FresnelDeformable.

tilt of the thumb, index-finger, and middle-finger contact surfaces within a range of -45° to 45° .

When the user grasps the device, the distance between the opposing fingertip contact surfaces is set to 6.5 cm. Previous studies [3, 4] have confirmed that pseudo-stiffness can arise for tangible objects measuring approximately 5 cm to 7 cm. Except for the servo motors and electronic components, all parts of the device were fabricated using a 3D printer with PLA filament. The force sensor is powered by a 5 V supply and connected in series with a 10 k Ω resistor as part of a voltage divider. The voltage across the sensor is read by a 12-bit analog-to-digital converter on the M5 ATOM S3, allowing forces of up to about 10 N to be detected under these conditions.

3.3 Apparatus

All participants wore a Meta Quest 3 headset throughout the experiment. Additionally, to mask any operational noise from the device, they wore headphones playing white noise. A virtual desk at the same height as the real one was presented within the virtual environment. Participants held the experimental device in their right hand and used their left hand to operate a keyboard for questionnaire responses.

3.4 Participants

We recruited ten participants, ranging in age from 24 to 48 years (Mean = 28.9, SD = 7.2; Male = 8, Female = 1, Non-binary = 1). All participants were right-handed and had prior experience with head-mounted display (HMD) based virtual reality. They were informed that the experiment would involve evaluating pseudo-stiffness in a VR environment. The total duration of the experiment did not exceed 20 minutes for any participant.

3.5 Procedure

Before the experiment began, participants were briefed on how to operate the device. They then practiced applying a force of 10 N using the actual device to prevent excessive force input during the experiment. After explaining the overall procedure and the number of conditions, participants donned the HMD and headphones, and the experiment commenced. Upon completion, they filled out a questionnaire and received a small confection as a token of appreciation.

3.6 Task

Participants freely interacted with each virtual object—defined by a combination of visual and haptic stimuli—for 10 seconds and then answered three questions:

- How soft did the object you grasped feel?
- Did the softness feel realistic?

- Were the visual and tactile stimuli coherent?

Responses were entered via a slider controlled by the arrow keys on a keyboard. At the leftmost end of the slider, the labels read “Not soft at all,” “Not realistic at all,” and “Incoherency,” respectively. At the rightmost end, the labels read “Extremely soft,” “Extremely realistic,” and “Coherency.” Although participants responded by visually positioning the slider, the recorded data were integer values ranging from 0 to 100.

After the experiment, participants completed a final questionnaire in which we measured the length of each participant’s hand (from the wrist to the tip of the middle finger) and the width of the middle fingertip. They were also asked to rate their fatigue level and task difficulty on a seven-point Likert scale, and to provide any additional comments or feedback about the experiment.

3.7 Conditions

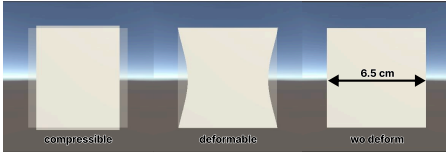


Figure 3: Apparent shape of each visual stimulus at maximum deformation.

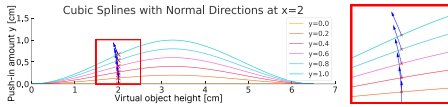


Figure 4: Cubic spline curves for each indentation value y [cm] used to generate the visual and tactile stimuli.

We prepared a total of six experimental conditions by combining three levels of visual stimuli (compressible, deformable, and wo/ deform) with two levels of tactile stimuli (with and without tilt). Table 1 summarizes these conditions, and Figure 3 shows the maximum visual deformation under each visual stimulus.

Table 1: Experimental conditions combining visual and tactile stimuli.

	Visual		
	compressible	deformable	wo/ deform
Tactile (wo/ tilt)	C	D	W
Tactile (w/ tilt)	CT	DT	WT

For the visual stimulus, three conditions were prepared: *Compressible*, *Deformable*, and *wo/ deform*. Figure 3 provides an overview of each condition. In the *Compressible* condition, the deformation was identical to that used in previous pseudo-stiffness research [4]. The *Deformable* condition was defined via a cubic spline curve, whose central control point moves according to the amount of indentation. Figure 4 illustrates the cubic spline curve applied for

object deformation as well as the normal direction of the contact surface for the index and middle finger. Finally, in the *wo/ deform* condition, the virtual object showed no visual deformation regardless of the applied force. In all three visual conditions, a force input of 10 N resulted in a deformation of 1 cm, corresponding to 15.4% of the baseline length of 6.5 cm.

The tactile stimulus was determined such that, in the *Deformable* condition, the contact normal and the visual deformation would align. Let y be the indentation proportional to the applied force, and let d be the change in angle. These are related by $d = 23.5 y$. This approximation achieves an accuracy of $R^2 = 0.993$, closely matching the normal direction of the visually deformed object.

3.8 Experiment Design

A within-subjects design was employed in this study. Each participant experienced all combinations of the three visual stimuli (compressible, deformable, and wo/deform) and the two tactile stimuli (with and without tilt), in three separate blocks. For each trial, participants answered three questions (coherency, realism, and softness). Thus, each participant completed a total of 18 trials, providing 54 responses. The order of conditions was determined using a predefined Latin square.

3.9 Results

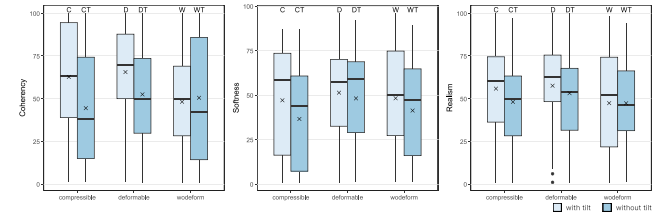


Figure 5: Boxplots of the measurement results for Coherency, Realism, and Softness.

Figure 5 presents boxplots of the experimental results. We analyzed these data with a two-way ANOVA (visual factor vs. tactile factor) using Coherency, Realism, and Softness as dependent variables. Where the Shapiro–Wilk test indicated non-normality, an aligned rank transform (ART [15]) was applied prior to the ANOVA.

3.10 Coherency

For Coherency, we applied an aligned rank transform (ART) ANOVA due to non-normality detected by the Shapiro–Wilk test in the C, CT, and D conditions. The main effect of the visual factor then showed a trend toward significance, $F(2, 18) = 2.90$, $p = 0.081$, partial $\eta^2 = 0.244$, indicating a medium effect size. The main effect of the tactile factor was significant, $F(1, 9) = 6.45$, $p = 0.032$, partial $\eta^2 = 0.418$, suggesting a large effect size. No interaction was found between the visual and tactile factors, $F(2, 18) = 1.41$, $p = 0.269$, partial $\eta^2 = 0.136$.

Post hoc comparisons (ART-C [6]) revealed a trend toward significance between the wo/ deform and deformable conditions for the visual factor (contrast estimate = 15.9, SE = 6.64, $t(18) = 2.40$, $p = 0.083$). For the tactile factor, a significant difference was found

(contrast estimate = 14.1, SE = 5.55, $t(9) = 2.54$, $p = 0.032$, Cohen's $d = 1.94$).

3.11 Realism

For Realism, we applied a conventional two-way ANOVA without transformation, as the Shapiro–Wilk test confirmed normality in all conditions. No significant main effect of the visual factor was found, although a medium effect size was suggested, $F(2, 18) = 2.09$, $p = 0.152$, partial $\eta^2 = 0.189$. The tactile factor also showed no significant main effect, but again with a medium effect size, $F(1, 9) = 1.94$, $p = 0.197$, partial $\eta^2 = 0.177$. No interaction was detected, and the effect size was small, $F(2, 18) = 0.29$, $p = 0.752$, partial $\eta^2 = 0.031$.

3.12 Softness

For Softness, we applied an aligned rank transform (ART) ANOVA due to non-normality detected in the C, CT, D, DT, and W conditions by the Shapiro–Wilk test. No significant main effect of the visual factor was found, $F(2, 18) = 0.91$, $p = 0.419$, partial $\eta^2 = 0.092$, indicating a small effect size. The tactile factor exhibited a trend toward significance with a medium-to-large effect size, $F(1, 9) = 3.90$, $p = 0.080$, partial $\eta^2 = 0.302$. No interaction was observed, $F(2, 18) = 0.40$, $p = 0.673$, partial $\eta^2 = 0.043$, also reflecting a small effect size.

Post hoc comparisons (ART-C) revealed a trend toward significance for the tactile factor (contrast estimate = 13.4, SE = 6.78, $t(9) = 1.98$, $p = 0.080$, Cohen's $d = 1.65$).

4 Discussion

We initially hypothesized (H1) that manipulating the tilt of each fingertip contact surface would allow the perception of softness. Our findings partially support this hypothesis. For the Softness metric, the tactile factor showed a trend toward significance with a medium-to-large effect size, suggesting that this factor influenced perceived softness independently of visual deformation. Neither a main effect of the visual factor nor an interaction between visual and tactile factors was observed. This aligns with previous research [4], which reported that a stiffness of at least 24 N/cm is necessary for perceiving pseudo-stiffness, suggesting that purely visual deformations may not be effective under certain stiffness conditions.

Our second hypothesis (H2) posited that manipulating the tilt of each fingertip contact surface would enhance the presentation of softness via pseudo-stiffness. However, this hypothesis was not supported. Although the Coherency metric showed that the tactile factor significantly improved visuo-tactile coherency, we did not observe any interaction effects between the visual and tactile factors in any of the measured indices. Consequently, no direct link was found between improved visuo-tactile coherence and the resulting perception of softness.

5 Limitation and Future Work

The device used in this study supports grasping interactions only with three fingers (thumb, index, and middle). Moreover, because the input interface relies solely on buttons mounted on the microcontroller, it is challenging to directly apply the current device

design to diverse VR applications. These interaction limitations stemming from the hardware configuration must be addressed through future improvements in the device.

With respect to tilt manipulation, we used a fixed maximum deformation in the present work. However, accommodating the wide variety of possible deformations in virtual objects will require a dynamic calculation of tilt angles. By implementing a system that combines mesh deformation based on physical simulations with an acquisition of mesh normals for tilt presentation, it would become feasible to develop a broader range of applications. In addition, we did not control the orientation of the thumb's contact surface in this experiment.

Achieving visually realistic objects presents another important challenge, aligned with the aforementioned limitations. We believe our proposed method could be extended to interactive experiences involving biological materials or foods. Improving the system to support such complex and realistic interactions remains a key direction for future work.

6 Conclusion

In this study, we investigated how combining visual deformation (pseudo-stiffness) with dynamic tilt manipulation of fingertip contact surfaces influences the perception of softness. Our results showed that controlling the tilt of each fingertip contact surface significantly improved visuo-tactile coherence and exhibited a medium-to-large effect size on perceived softness. However, no interaction effects were found between the visual and tactile factors, and the overall improvement in softness perception was limited. These findings imply that while purely tactile cues from fingertip tilt manipulation can enhance the user's sense of consistency between vision and touch, further refinement is needed to exploit the full potential of visuo-tactile integration for conveying a broader range of softness.

Ultimately, we seek to establish a more versatile framework for designing haptic interfaces that effectively render both rigid and highly compliant virtual objects.

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